

Linear Regression:

Prediction function:

$$\hat{y}_i = \beta_0 + \sum_{j=1}^n \beta_j x_{ij}$$

- $\hat{y}_i$ = predicted value
 - β_0 = intercept (bias term)
 - β_j = coefficient for feature j
 - x_{ij} = feature j of sample i
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Residual (error):

$$e_i = y_i - \hat{y}_i$$

- e_i = prediction error
 - y_i = actual target value
 - $\hat{y}_i$ = predicted target
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Cost function (MSE):

$$J(\beta) = \frac{1}{2N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- $J(\beta)$ = loss to minimize
 - N = number of samples
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Gradient of cost wrt coefficient:

$$\frac{\partial J}{\partial \beta_j} = -\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i) x_{ij}$$

- Measures slope of cost wrt parameter β_j
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Gradient descent update rule:

$$\beta_j \leftarrow \beta_j - \alpha \frac{\partial J}{\partial \beta_j}$$

- α = learning rate (step size)
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Normal equation (closed form):

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

- X = feature matrix
 - y = target vector
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Regularization:

- Ridge (L2):

$$J(\beta) = \frac{1}{2N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

- Lasso (L1):

$$J(\beta) = \frac{1}{2N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n |\beta_j|$$

- λ = penalty term controlling shrinkage
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Evaluation metrics:

- $MSE = \frac{1}{N} \sum (y_i - \hat{y}_i)^2$
 - $RMSE = \sqrt{MSE}$
 - $MAE = \frac{1}{N} \sum |y_i - \hat{y}_i|$
 - $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$
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Logistic Regression:

Sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Maps any real z to $(0, 1)$
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Linear input to sigmoid:

$$z_i = \beta_0 + \sum_{j=1}^n \beta_j x_{ij}$$

- z_i = linear combination of features
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Prediction (probability):

$$p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}$$

- p_i = probability that sample i belongs to class 1
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Decision rule:

$$\hat{y}_i = \begin{cases} 1 & \text{if } p_i \geq 0.5 \\ 0 & \text{if } p_i < 0.5 \end{cases}$$

Logit (log-odds):

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \sum_{j=1}^n \beta_j x_{ij}$$

- Connects probability to linear model
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Cost function (log loss):

$$J(\beta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

- Penalizes incorrect probability predictions
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Gradient of cost wrt coefficient:

$$\frac{\partial J}{\partial \beta_j} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i) x_{ij}$$

Update rule:

$$\beta_j \leftarrow \beta_j - \alpha \frac{\partial J}{\partial \beta_j}$$

Regularization:

- Ridge:

$$J(\beta) = \text{LogLoss} + \lambda \sum_{j=1}^n \beta_j^2$$

- Lasso:

$$J(\beta) = \text{LogLoss} + \lambda \sum_{j=1}^n |\beta_j|$$

Confusion matrix metrics:

- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
 - Precision = $\frac{TP}{TP+FP}$
 - Recall = $\frac{TP}{TP+FN}$
 - F1 = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
 - FPR = $\frac{FP}{FP+TN}$
 - TPR = $\frac{TP}{TP+FN}$
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Decision Trees:

Entropy (classification impurity):

$$H(S) = - \sum_{c=1}^C p_c \log_2(p_c), \quad p_c = \frac{|S_c|}{|S|}$$

- $H(S)$ = entropy of set S
 - p_c = proportion of class c in S
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Gini impurity:

$$Gini(S) = 1 - \sum_{c=1}^C p_c^2$$

Information gain (feature A):

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

- Measures impurity reduction after split
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Variance (regression trees):

$$Var(S) = \frac{1}{|S|} \sum_{i=1}^{|S|} (y_i - \bar{y}_S)^2$$

- $\bar{y}_S$ = mean target value in node
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Prediction rule:

- Classification $\rightarrow$ majority class in leaf
 - Regression $\rightarrow$ mean target in leaf
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Pruning (cost-complexity):

$$R_\alpha(T) = R(T) + \alpha|T|$$

- $R(T)$ = error of tree
 - α = penalty for number of leaves
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